

THE DEBUGGERS · SOFTWARE ENGINEERING

SecondHelping

THIS WEEKS PROGRESS REPORT

Campus Surplus-Food Marketplace · Oct 13–19, 2026

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WEEK · OVERVIEW

This week transformed the prototype.

A real payment lifecycle, a normalized food catalog, a rebuilt UI structure, and a published AI roadmap.

FIGURE 1 · this weeks MILESTONE TIMELINE — OCT 13 → OCT 21



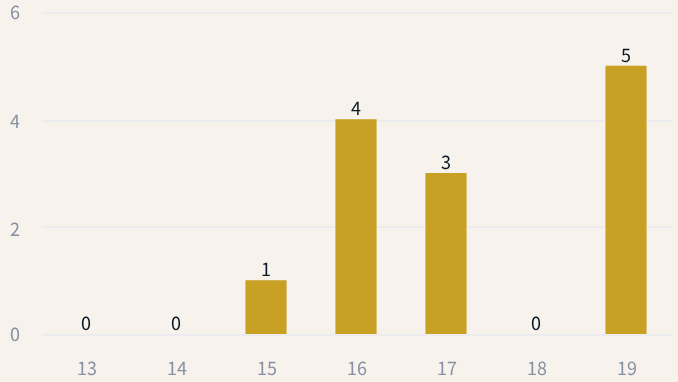
Six milestones across the week — from the ER draft on Oct 15 to the SRS v0.2 update on Oct 19.

Source: GitHub · bhsei/26C, git log & docs/database, Oct 13–21 2026

WEEK 5 · ACTIVITY

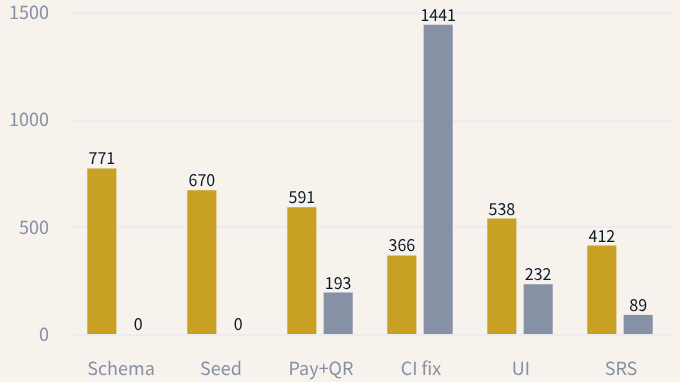
13 commits, five contributors, one sustained push.

FIGURE 2 · COMMITS PER DAY — OCT 13–19



Gold = added · slate = removed · net +1,477 lines, incl. 1,441-line cleanup

FIGURE 3 · LINES ADDED VS REMOVED



Commit of Oct 17 removed draft schema + seed files (−1,441 lines)

13	5	4	38
COMMITS	ACTIVE CONTRIBUTORS	NEW MIGRATIONS	FILES CHANGED

01 · DATABASE

From flat listings to a normalized food catalog.

Food becomes a first-class entity · orders carry a payment lifecycle · four new migrations

Listings now reference a food catalog; orders carry a payment lifecycle.

FIGURE 4 · BEFORE / AFTER — DATA MODEL



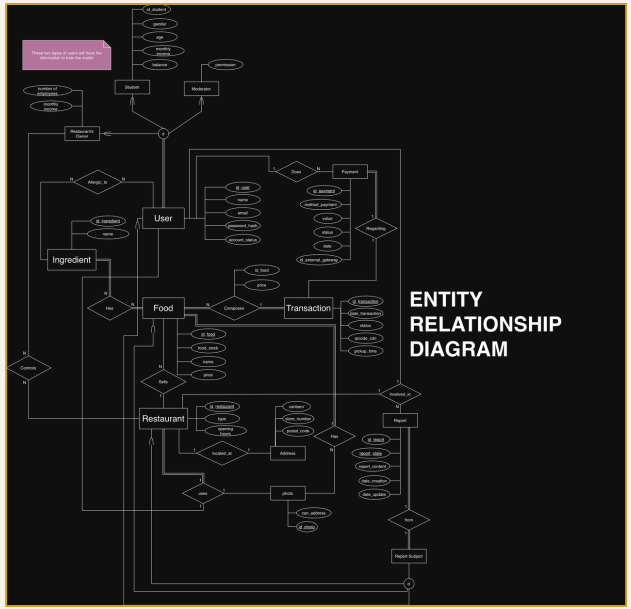
TABLE 1 · NEW MIGRATIONS — OCT 16–19

DATE	MIGRATION	PURPOSE
10-16	add_pending_payment_status	New PENDING_PAYMENT order status
10-16	add_payment_fields	paymentRef + paidAt on Order
10-17	add_food_model	Food table; Listing links via foodId
10-19	add_order_refunds	refundedAt + refundAmountCents

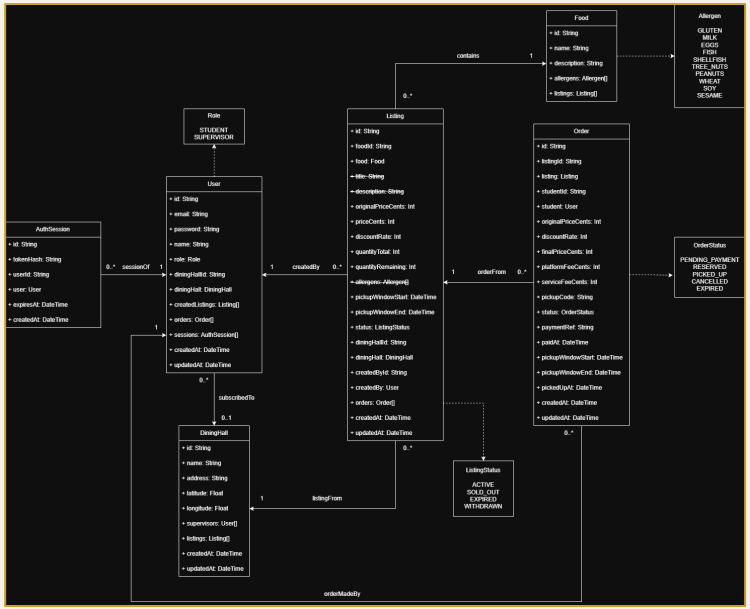
Source: prisma/schema.prisma · prisma/migrations/

Two drafts, one week: from concept to Prisma-faithful model.

FIGURE 5 · ER DIAGRAM EVOLUTION — v1 (OCT 3–12) → v2 (OCT 15–17)

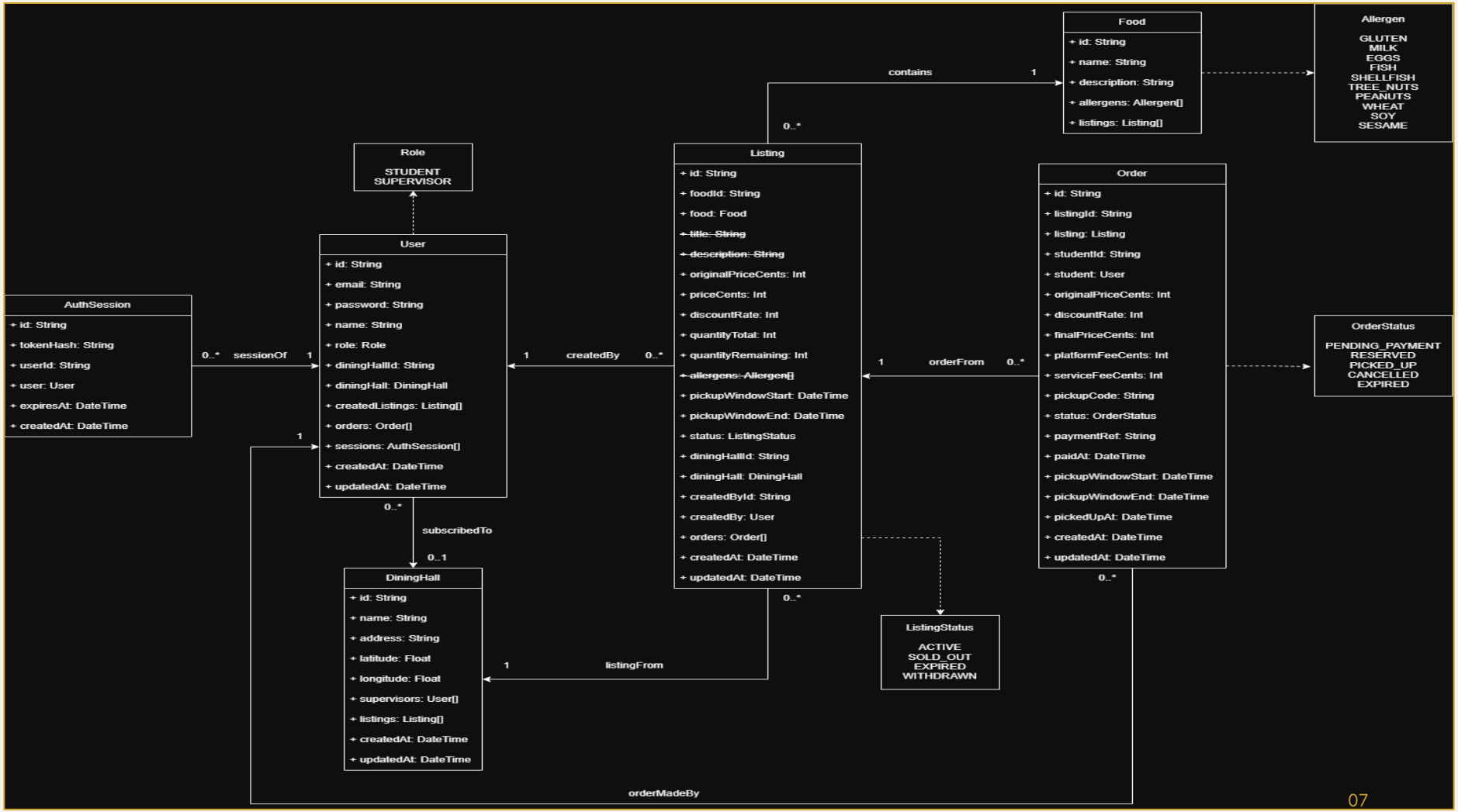


v1 · conceptual draft · Gabriel



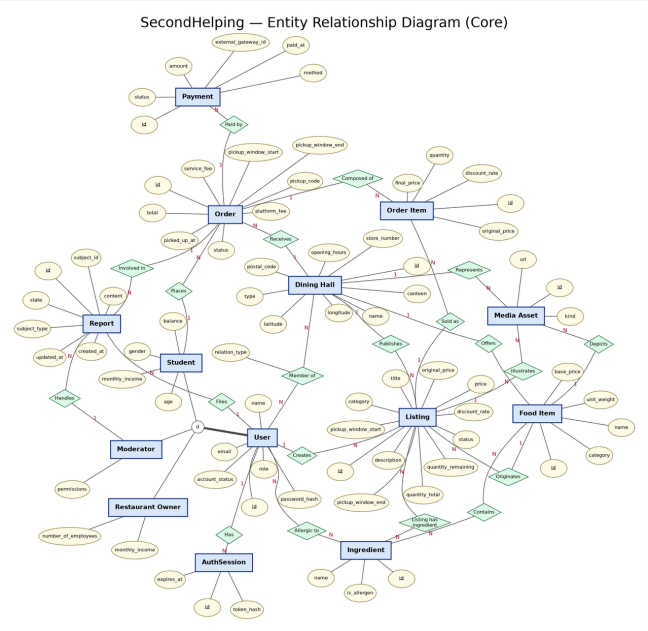
v2 · Prisma-faithful · Gabriel, Oct 15–17

Source: repo · docs/database/ · Old vs Entity Relationship Diagram (2026_10_17).png

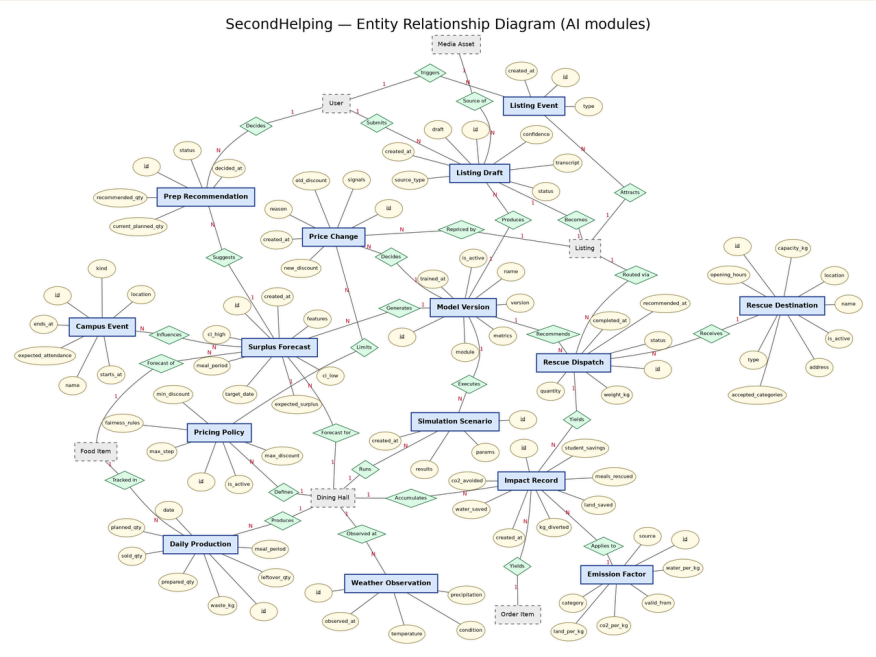

Food
+ id: String
+ name: String
+ description: String
+ allergens: Allergen[]
+ listings: Listing[]
Allergen
GLUTEN
MILK
EGGS
FISH
SHELLFISH
TREE_NUTS
PEANUTS
WHEAT
SOY
SESAME
contains
1
0..*
Role
STUDENT
SUPERVISOR
Listing
+ id: String
+ foodId: String
+ food: Food
+ title: String
+ description: String
+ originalPriceCents: Int
+ priceCents: Int
+ discountRate: Int
+ quantityTotal: Int
+ quantityRemaining: Int
+ allergens: Allergen[]
+ pickupWindowStart: DateTime
+ pickupWindowEnd: DateTime
+ status: ListingStatus
+ diningHallId: String
+ diningHall: DiningHall
+ createdById: String
+ createdBy: User
+ orders: Order[]
+ createdAt: DateTime
+ updatedAt: DateTime
Order
+ id: String
+ listingId: String
+ listing: Listing
+ studentId: String
+ student: User
+ originalPriceCents: Int
+ discountRate: Int
+ finalPriceCents: Int
+ platformFeeCents: Int
+ serviceFeeCents: Int
+ pickupCode: String
+ status: OrderStatus
+ paymentRef: String
+ paidAt: DateTime
+ pickupWindowStart: DateTime
+ pickupWindowEnd: DateTime
+ pickedUpAt: DateTime
+ createdAt: DateTime
+ updatedAt: DateTime
OrderStatus
PENDING_PAYMENT
RESERVED
PICKED_UP
CANCELLED
EXPIRED
User
+ id: String
+ email: String
+ password: String
+ name: String
+ role: Role
+ diningHallId: String
+ diningHall: DiningHall
+ createdListings: Listing[]
+ orders: Order[]
+ sessions: AuthSession[]
+ createdAt: DateTime
+ updatedAt: DateTime
AuthSession
+ id: String
+ tokenHash: String
+ userId: String
+ user: User
+ expiresAt: DateTime
+ createdAt: DateTime
0..*
sessionOf
1
1
createdBy
0..*
1
orderFrom
0..*
1
0..*
subscribedTo
0..1
DiningHall
+ id: String
+ name: String
+ address: String
+ latitude: Float
+ longitude: Float
+ supervisors: User[]
+ listings: Listing[]
+ createdAt: DateTime
+ updatedAt: DateTime
1
listingFrom
0..*
ListingStatus
ACTIVE
SOLD_OUT
EXPIRED
WITHDRAWN
0..*
orderMadeBy

Next Week - Implement the AI-Module's data structure onto PRISMA

FIGURE 6 · ER DIAGRAM CORE AND AI-MODULES — (OCT 16-19)



v1 · conceptual draft · Gabriel



v2 · Prisma-faithful · Gabriel, Oct 15–17

Source: repo · docs/database/ · Old vs Entity Relationship Diagram (2026_10_17).png

02 · PAYMENT & ORDERS

Reserving a meal now means paying for it — a WeChat-style flow, simulated end to end.

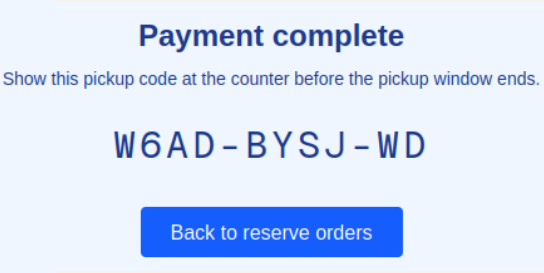
Claim → pay → reserved · pickup code issued · platform and service fees modeled

Claim ›› pay ›› reserved: a simulated WeChat payment lifecycle.

FIGURE 6 · PAYMENT SEQUENCE & PICKUP CODES

1. Student claims a listing — order created as PENDING_PAYMENT
2. Redirect to /pay/[ref] — the order's payment page
3. FakeQR renders a WeChat-style QR with amount + fees
4. Student confirms — confirmPayment action runs
5. Order flips to RESERVED — pickup code issued (XXXX-XXXX-XX)

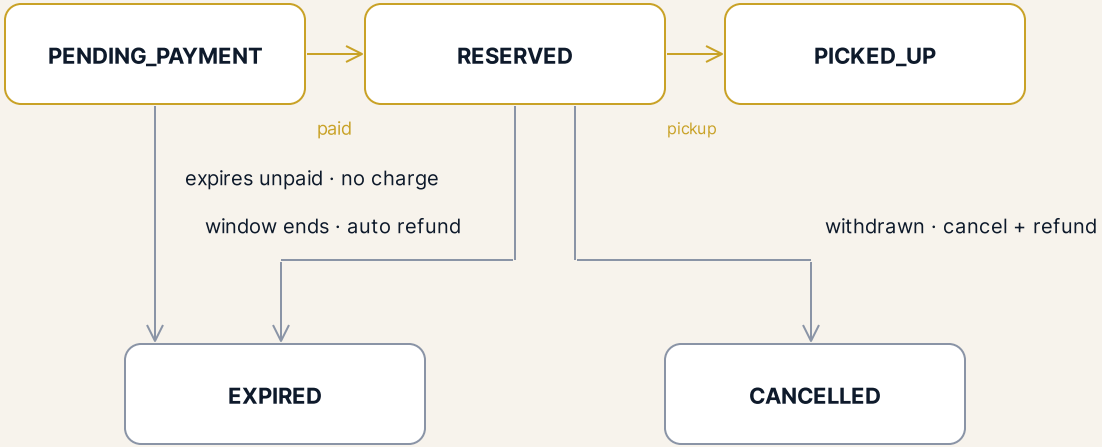




Source: app/actions.ts · app/pay/[ref] · prisma/seed.ts

Every order is settled automatically — refunds included.

FIGURE 7 · ORDER STATE MACHINE — lib/order-lifecycle.ts



Full refund = meal price × quantity + platform fee + service fee

01 WINDOW EXPIRY

Listings flip to EXPIRED when the pickup window ends

02 AUTO REFUND

Paid orders refund the meal plus both fees

03 WITHDRAWALS

Withdrawn listings cancel orders and refund

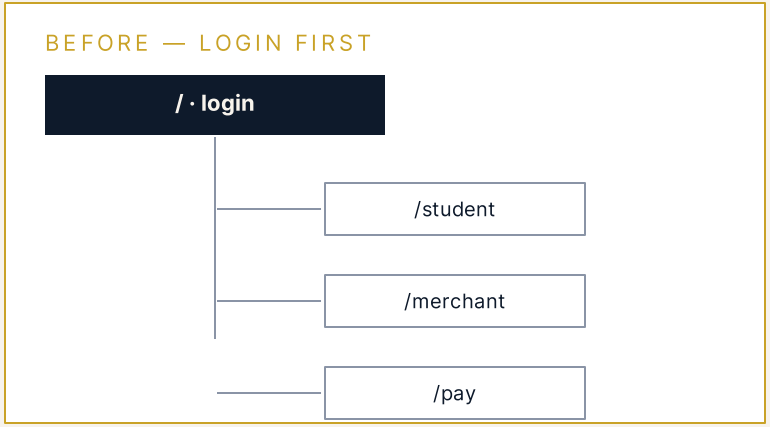
04 SAFE UPDATES

Conditional updateMany keeps concurrency safe

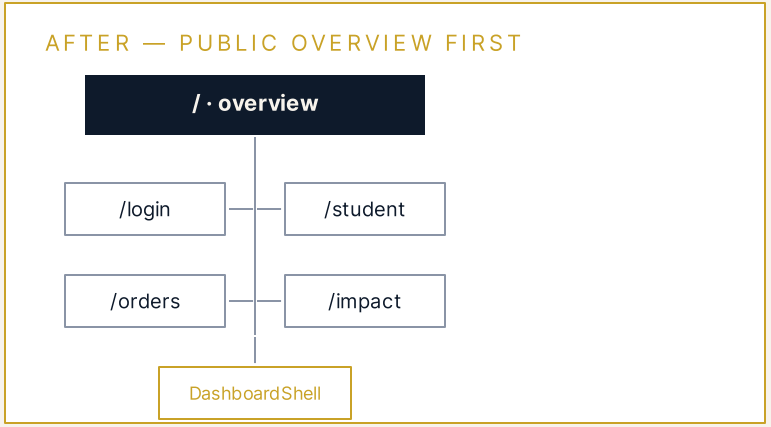
Source: lib/order-lifecycle.ts · settleUnavailableOrders

Public first — the student page shrank to five lines.

FIGURE 8 · SITEMAP BEFORE / AFTER



4 routes · every page behind login



public overview before login · shared shell

01 **PUBLIC OVERVIEW**

The / route is an overview page — no login wall

02 **STUDENT PAGE**

Slimmed 132 → 5 lines via component extraction

Source: app/ routes · student page measured via git

Supervisors pick from the catalog; validation guards every listing.

Publishing now runs the full gauntlet: catalog pick, price, discount, quantity, window, duplicates.

TABLE 2 · LISTING PUBLICATION VALIDATION

FIELD	RULE	FAILURE BEHAVIOR
foodId	Must exist in Food catalog	Reject with catalog error
price	originalPriceCents > 0	Reject invalid price
discount	Integer percent, 0–100	Reject out of range
quantity	quantityTotal > 0	Reject non-positive
window	Start > now • end > start	Reject past window
duplicates	Same food + hall, ACTIVE & unexpired	Block duplicate listing

01 **SERVER-SIDE GUARD**

A bad listing never reaches the storefront — validation runs before anything is stored.

02 **FUTURE-FIRST WINDOWS**

Pickup windows must lie in the future; discounts are stored as integer percent.

Source: app/actions.ts · createListing

The spec grew ~400 lines — AI is now on the roadmap.

TABLE 3 · SRS v0.2 — AI/ML REQUIREMENTS · REQ-FUNC-009 → 014

ID	REQUIREMENT	ONE-LINE DESCRIPTION
REQ-FUNC-009	Predictive Surplus	Forecast leftover meals from demand patterns
REQ-FUNC-010	Prep Copilot	Suggest prep quantities via a digital twin
REQ-FUNC-011	Multimodal Listing	Create listings from photo, voice or text
REQ-FUNC-012	Dynamic Pricing	Adjust discounts automatically by demand
REQ-FUNC-013	Rescue Routing	Route surplus to nearby takers at window end
REQ-FUNC-014	Impact Estimator	Quantify food saved, CO_2 and cost avoided

APPENDIX C · AI ROADMAP

SENSE | PREDICT | PREVENT | PRICE | PERSONALIZE | PICKUP | PROVE | LEARN

All six AI/ML requirements marked Planned in the new 3.6 section · docs/SRS.md grew to 665 lines

Source: docs/SRS.md · §3.6 AI/ML + Appendix C

CI is green again — Postgres 16, Prisma 7, frozen deps.

01 **POSTGRES 16 SERVICE**

CI spins up Postgres 16 on 5432 with a healthcheck

02 **PRISMA 7 CONFIG**

prisma7.config.ts replaces the classic setup

03 **FROZEN LOCKFILE**

pnpm install --frozen-lockfile keeps deps reproducible

04 **LINT + BUILD GATE**

Lint and build must pass — 15-minute job budget

TABLE 4 · CI PIPELINE STAGES

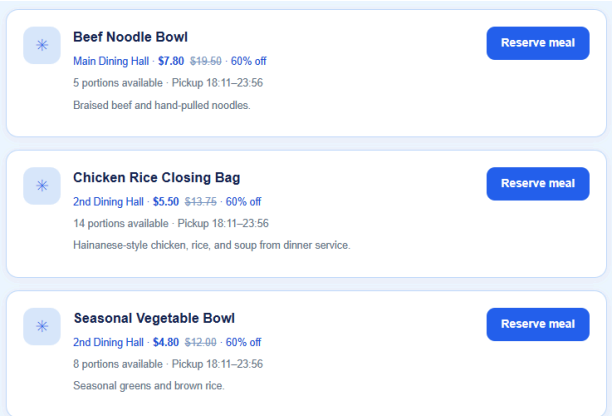
STAGE	ACTION
checkout	fetch repository
pnpm	install --frozen-lockfile
generate	prisma generate
validate	prisma validate
migrate	prisma migrate deploy
lint	eslint gate
build	production build

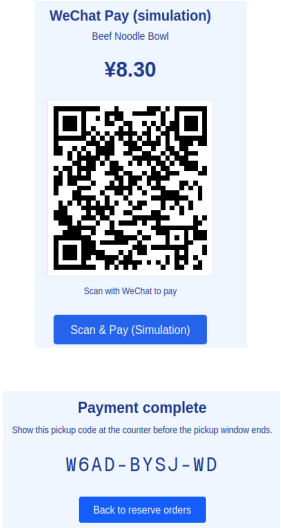
Source: .github/workflows/ci.yml · prisma7.config.ts

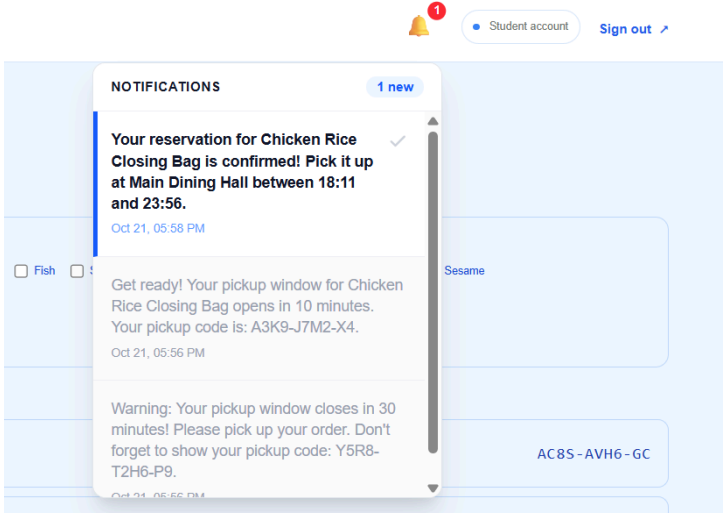
Student account

Reservation confirmation

1 — 2 — 3

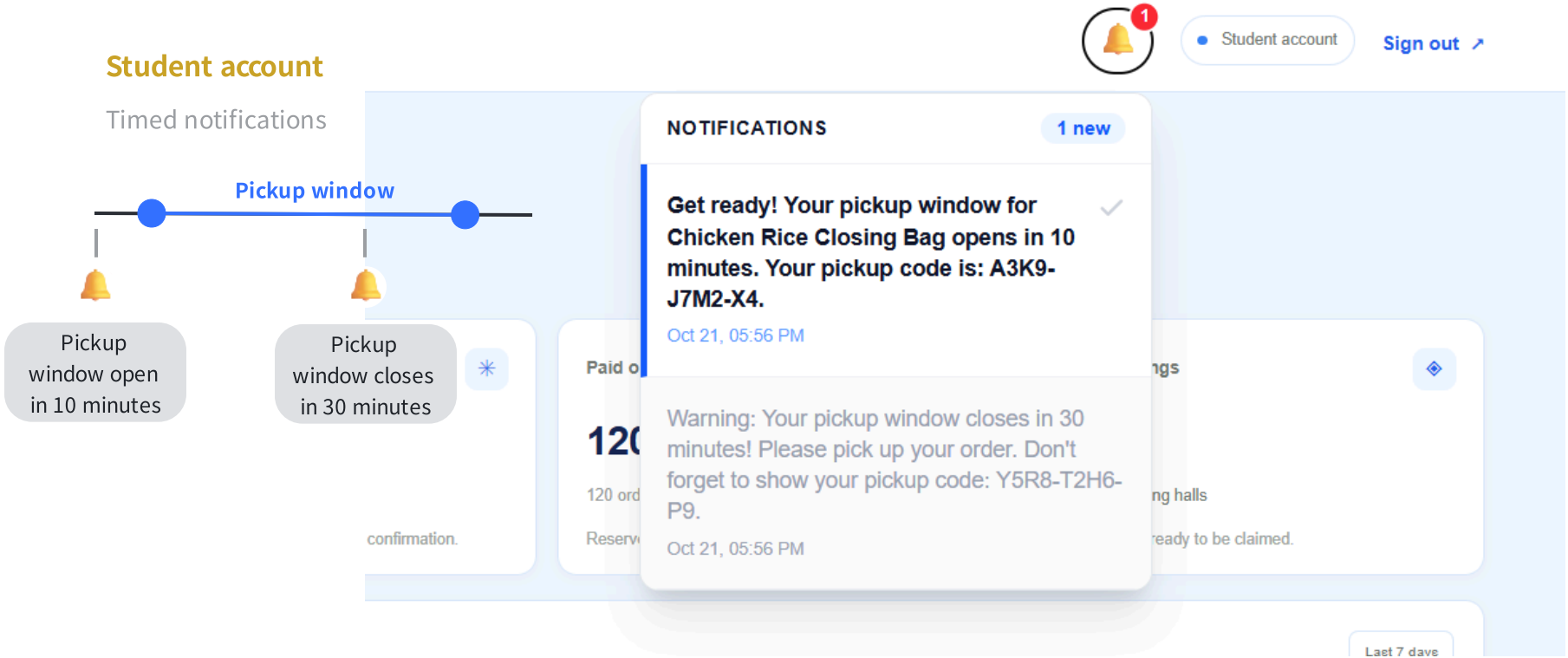






Student account

Timed notifications



The prototype keeps moving: pickup, cancellation, impact.

01

REAL QR PICKUP

Scan pickup codes at the counter — staff confirms fulfilment in seconds.

02

CANCELLATION + STOCK RELEASE

Orders can be cancelled; quantity returns to the listing for the next taker.

03

IMPACT ESTIMATOR v0

First version of the food-saved and CO_2 estimator — the REQ-FUNC-014 seed.

04

RULE-BASED PRICING

Discount rules keyed to demand and window proximity — a first step toward dynamic pricing.

Source: docs/SRS.md · REQ-FUNC-009 → 014



Thank you.

Questions & discussion welcome

github.com/bhsei/26C